

MASTER'S COMPREHENSIVE EXAMINATION

Part I Theory (2 hours) April 2nd, 2016

Instructions: Attempt any three out of the following four questions on both pages.
Mention which three you would like to have graded. Open book and open notes.

1. (a) The random variable X has the property that all its moments are equal; more specifically, $E(X^n) = c$ for all integers $n \geq 1$ and some constant c , $0 < c < 1$. Find the distribution of X .

(b) For any random variable Y show that

$$E(Y) = \int_0^{\infty} P(Y > y) dy - \int_0^{\infty} P(Y > -y) dy.$$

2. Let X_1, X_2, X_3 be independent $N(\mu, \sigma^2)$, μ and σ^2 unknown.
Find a 98% CI for σ^2 given $X_1 = 1, X_2 = 0, X_3 = -1$.

3. Let X be a random sample of size 1 have an exponential distribution with parameter θ .
That is, $f_x(x; \theta) = \frac{1}{\theta} e^{-x/\theta}$, $x > 0$, $\theta > 0$.

Find the most powerful test of size e^{-1} for $H_0: \theta = 1$ vs $H_1: \theta = 3$.

4. (a) Consider a point on a grid that moves to a new position in n steps. At each step, the point can move up one unit with probability p or right one unit with probability $1 - p$. What is the expected area of the smallest rectangle containing the starting and ending positions of the point?

(b) A point is chosen uniformly at random on a line segment of length L . Let Y be the ratio of the length of the shorter segment to the length of the longer segment determined by the selected point. Find the cumulative distribution function of Y .